Infosys Springboard Virtual Internship Batch-11

Food Trends Understanding Customer Preferences

Team-D

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Contents:

1. Project Objective

The primary objective of this project is to design and implement an end-to-end analytical dashboard using Power BI that enables stakeholders to understand, monitor, and optimise the performance of a food delivery ecosystem.

This project specifically aims to:

- Analyse business performance by tracking key metrics such as total revenue, total orders, average order value, and restaurant coverage.
- Understand customer behaviour by studying demographic factors, including age, gender, income levels, occupation, and family size, and their influence on ordering patterns.
- Evaluate restaurant performance using indicators such as ratings, order volumes, cuisine offerings, and revenue contribution.
- Identify food and menu trends by analysing cuisine popularity, veg vs non-veg preferences, top-selling items, pricing distribution, and revenue-generating dishes.
- Examine ordering dynamics such as order frequency, day-wise demand, revenue volatility, and long-term trends.
- Support data-driven decision-making by transforming raw transactional data into actionable insights that can guide marketing strategies, pricing decisions, menu optimisation, and customer retention initiatives.

2. Project Description

2.1 Overview

This project focuses on building a comprehensive analytical dashboard for a food delivery platform using Power BI. The dashboard integrates data from five interrelated datasets—customers, orders, restaurants, menu, and food items—to provide a 360-degree view of business operations.

The raw datasets contain high-volume transactional and master data with inconsistencies such as:

- o Duplicate records
- o Missing values
- o Multi-valued attributes (e.g., cuisines)
- o Mixed data formats (dates, currency, identifiers)
- o Through systematic data transformation, modeling, and visualization, the project converts raw data into meaningful insights that support:Strategic planning

- Customer behavior analysis
- Restaurant and menu optimization
- Revenue growth and retention strategies
- The final output is a 7-page interactive Power BI dashboard, each page addressing a specific business question.

2.2 Project Approach

The project followed a structured data analytics lifecycle, ensuring accuracy, scalability, and usability at every stage.

Step 1: Data Understanding

- Studied schema, column definitions, and business meaning of all five tables
- Identified primary keys, foreign keys, and potential relationships
- Understood domain context (food delivery operations)

Step 2: Data Transformation

- Cleaned and standardised datasets using Power Query
- Handled missing, duplicate, and inconsistent values
- Created derived columns to support analysis

Step 3: Data Modelling

- Established relationships across tables
- Designed an optimised data model to avoid redundancy
- Ensured correct filter propagation

Step 4: Dashboard Development

- Identified key KPIs and metrics
- Selected appropriate visuals for each insight
- Designed user-friendly layouts with slicers and filters

Step 5: Insight Validation

- Cross-checked measures across visuals
- Ensured business logic consistency
- Finalized insights and interpretations

2.3 Technology Used

Power BI Desktop

- Core tool for dashboard creation and reporting
- Used for visual design, DAX calculations, and interactivity

Power Query Editor

- Used for ETL (Extract, Transform, Load) operations
- Enabled data cleansing, transformation, and shaping

- DAX (Data Analysis Expressions)
- Created calculated measures such as:
  - Total Revenue
  - Average Order Value
  - Orders per Day
  - Ranking and segmentation metrics

CSV / Excel Data Sources
- Provided structured datasets for analysis
- Enabled real-world data handling experience

2.4 Insights from the Dashboard

The dashboard delivers insights across multiple dimensions:
- Business Performance – Revenue trends, city-wise contribution, order volume
- Customer Behaviour – Age, income, gender, occupation-based spending patterns
- Restaurant Efficiency – Ratings, order volume, revenue share
- Food Trends – Cuisine popularity, veg vs non-veg demand
- Ordering Dynamics – Frequency, volatility, peak days

These insights allow stakeholders to move from reactive decisions to proactive strategies.

2.5 Real-World Impact for Media and Public Communication

The insights generated through this dashboard can be effectively used for:
- Public reports on food consumption trends
- Media storytelling on customer preferences
- Business intelligence briefings
- Market research and competitive analysis
- The project demonstrates how analytics supports transparent, data-backed communication.

3. Timeline Overview

The project was executed over four clearly defined milestones, ensuring disciplined progress and quality control.

Milestone	**Activities Covered**	**Date**
Milestone 1	Data transformation and preprocessing	01/01/2026
Milestone 2	Dashboard Development-Pages 1,2,3	09/01/2026
Milestone 3	Dashboard Development-Pages 4,5	16/01/2026
Milestone 4	Dashboard Development-Pages 6,7	23/01/2026

Each milestone acted as a quality checkpoint, ensuring readiness before proceeding to the next phase.

4. Project Milestones

4a. Key Milestones

The project was executed through four well-defined milestones, each representing a distinct phase in the analytics workflow. These milestones ensured systematic progress from data preparation to advanced dashboard insights.

Milestone 1 – Data Transformation (01/01/2026)

This milestone focused on converting raw data into a clean, consistent, and analysis-ready format. All five datasets—customers, orders, restaurants, menu, and food—were processed using Power Query.

Key activities included:

- Removal of duplicate and null records to ensure data accuracy
- Standardisation of text fields such as city names, cuisine types, and categorical attributes
- Splitting and unpivoting of multi-valued columns (for example, cuisine fields)
- Creation of derived columns such as order day, income bounds, rating categories, and currency-normalised sales amounts
- Validation and correction of data types for numerical, categorical, and date fields

Outcome:
Five transformed and standardised datasets forming a strong foundation for dashboard development.

Milestone 2 – Dashboard Pages 1, 2, and 3 (09/01/2026)

This milestone involved the development of the core analytical dashboards, focusing on overall performance, customer behaviour, and restaurant analysis.

Key deliverables:

- Page 1: Business Overview – KPIs, revenue trends, and city-wise performance
- Page 2: Customer Analysis – demographics, income, and spending behaviour
- Page 3: Restaurant Performance – ratings, order volume, and revenue contribution

Outcome:
High-level dashboards providing immediate visibility into business health and customer patterns.

Milestone 3 – Dashboard Pages 4 and 5 (16/01/2026)

This milestone concentrated on product-level and operational insights, with emphasis on food items, menus, and ordering behaviour.

Key deliverables:

- Page 4: Food and Menu Trends – cuisine popularity, veg vs non-veg analysis, and pricing distribution
- Page 5: Ordering Insights – order frequency, demand volatility, and peak ordering patterns

Outcome:
Dashboards enabling tactical decisions related to menu planning and demand forecasting.

Milestone 4 – Dashboard Pages 6 and 7 (23/01/2026)

The final milestone focused on advanced analytics and strategic insights that support long-term planning.

Key deliverables:

- Page 6: Customer Demographics and Regional Analysis – age groups, city-level performance, and customer distribution
- Page 7: Loyalty, Revenue Drivers, and Strategic Trends – customer retention indicators and long-term growth insights

Outcome:

Strategic dashboards supporting executive-level decision-making and future planning.

4b. Project Execution Details

The execution of this project followed a systematic and structured workflow, ensuring accuracy, consistency, and meaningful insight generation at every stage. Each phase of execution aligned closely with the defined project milestones and contributed to the successful development of the 7-page Power BI dashboard.

Data Acquisition and Initial Setup

- The raw datasets were collected in CSV format and imported into Power BI Desktop.
- All five tables—customers, orders, restaurants, menu, and food—were reviewed to understand schema, data types, and business relevance.
- Column naming conventions and data consistency issues were identified prior to transformation.

Data Transformation and Cleaning (Milestone 1)

- Data cleaning was performed using Power Query Editor.
- Duplicate records and null values were removed to improve data reliability.
- Inconsistent text fields such as city names, cuisine types, and categorical attributes were standardised.
- Multi-valued fields (for example, cuisine columns) were split and unpivoted to enable accurate aggregation.
- Derived columns such as order day, income bound, rating categories, and currency-converted sales values were created.
- Unnecessary columns were removed to optimize performance.
- Data types were validated and auto-detected to ensure correct calculations.

Outcome: Clean, transformed, and analysis-ready datasets.

Data Modelling and Relationships

- Relationships between the five tables were established using appropriate keys.
- One-to-many and many-to-many relationships were carefully managed to avoid ambiguity.
- Cross-filter directions were validated to ensure correct data propagation across visuals.

- The data model was optimized to reduce redundancy and improve query performance.
- Outcome: A robust and scalable data model supporting accurate analytics.
- KPI Definition and DAX Measure Development
- Business requirements were translated into measurable KPIs.
- DAX measures were created for:
  - Total Revenue
  - Total Orders
  - Average Order Value
  - Orders per Day
  - Customer Segmentation Metrics
  - Cuisine and Restaurant Rankings

Measures were validated across multiple visuals to ensure consistency

Outcome: Reliable KPIs enabling meaningful business insights.

Dashboard Development – Milestone-wise Execution

Milestone 2: Pages 1, 2, and 3

Designed dashboards for:

- Overall Business Overview
- Customer Analysis
- Restaurant Performance
- Implemented KPI cards, bar charts, line charts, donut charts, and tables.
- Added slicers for city, cuisine, and date-based filtering.
- Ensured logical navigation and consistent visual formatting.

Milestone 3: Pages 4 and 5

Developed dashboards focused on:

- Food and Menu Trends
- Ordering Behavior and Demand Patterns
- Created visuals for:
  - Cuisine popularity
  - Veg vs Non-Veg comparison
  - Price and revenue distribution
  - Order frequency and volatility
  - Enabled drill-down analysis for deeper insights.

Milestone 4: Pages 6 and 7

Built dashboards for:

- Customer Demographics and Regional Insights
- Loyalty, Revenue Drivers, and Strategic Trends
- Integrated long-term trend analysis and segmentation-based insights.
- Ensured dashboards support strategic decision-making.

5. Snapshots / Screenshots

This section includes screenshots of all 7 dashboard pages, demonstrating:

- KPI cards
- Charts and tables
- Filters and slicers
- Interactive drill-downs

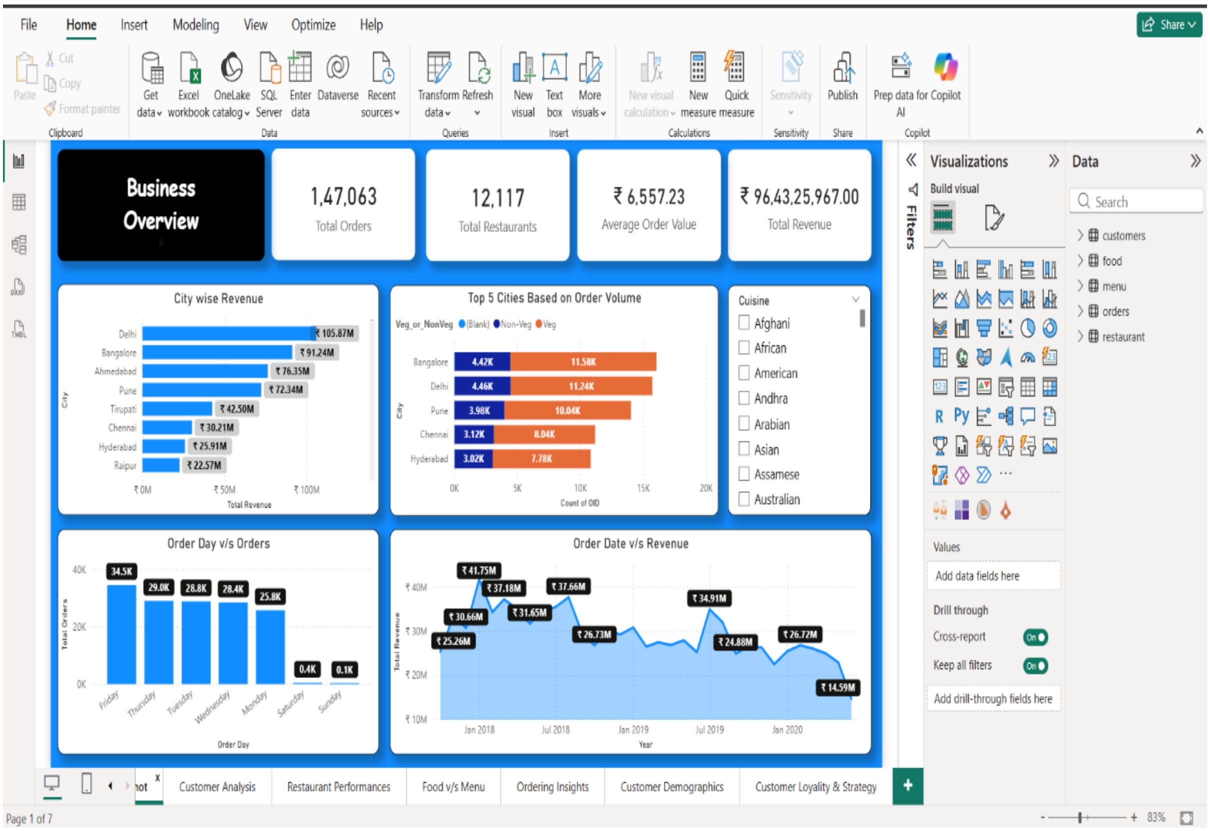
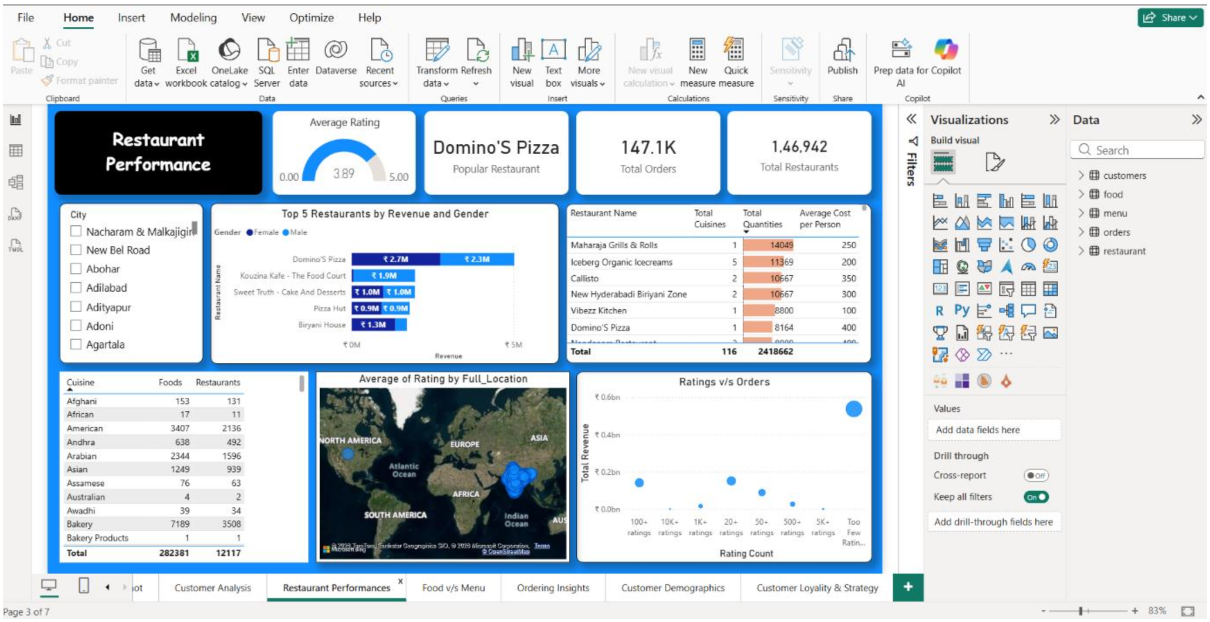
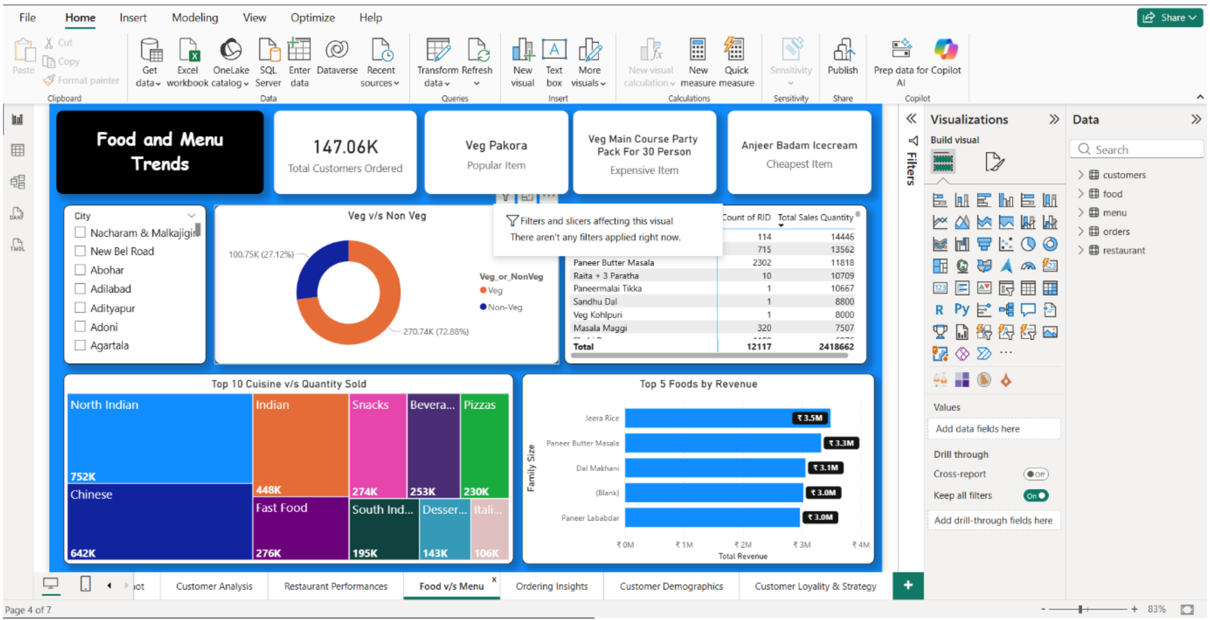


Fig1: Business Overview

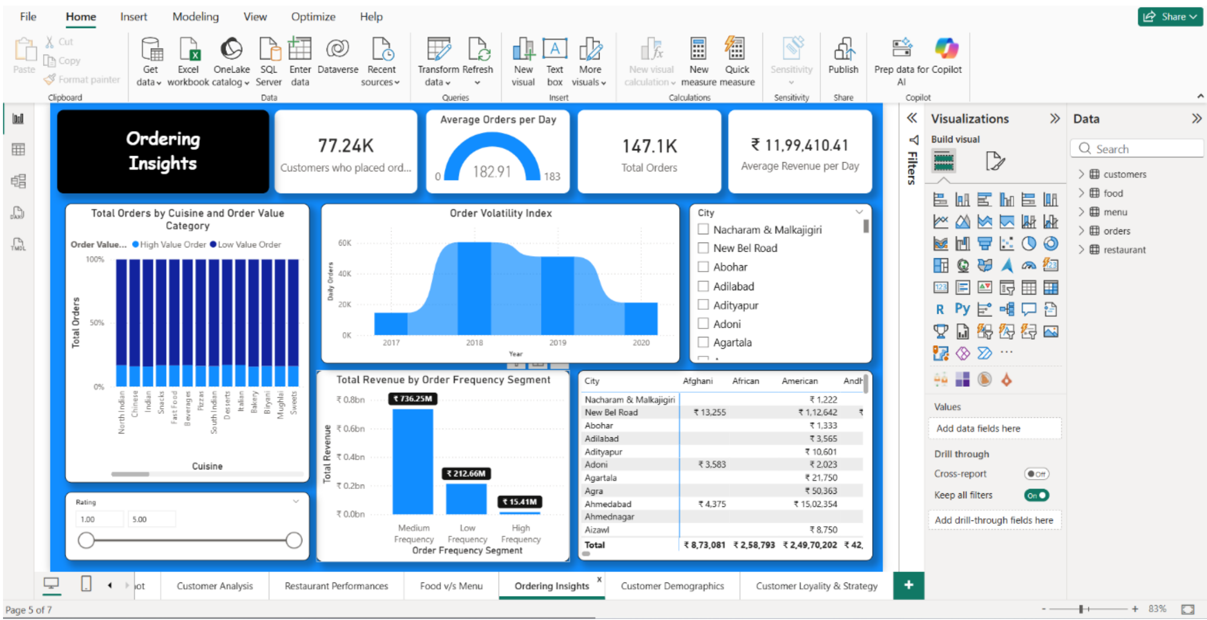
2: Customer Analysis



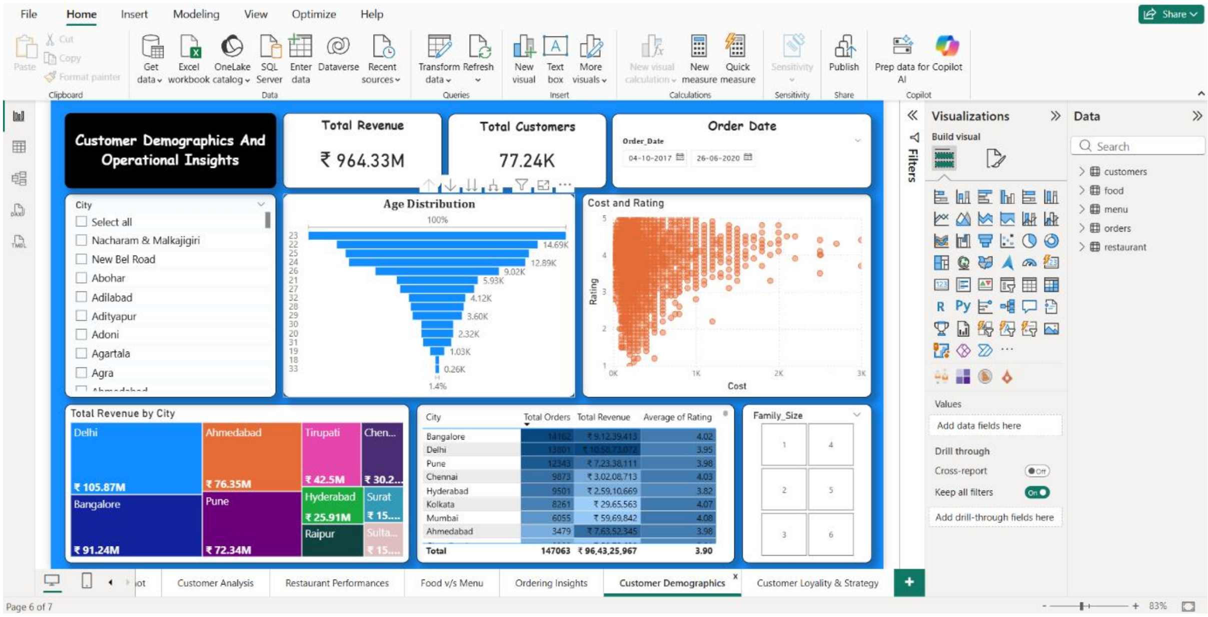
3:Restaurant Performance



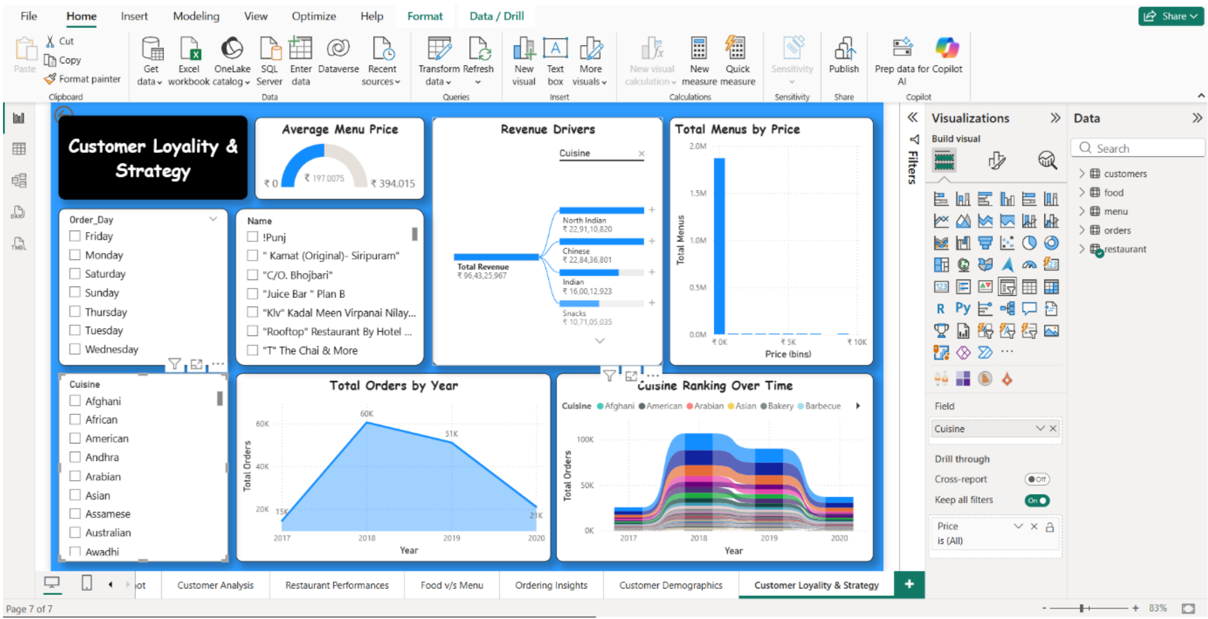
4:Food and Menu Trends



5:Ordering Insights



6:Customer Demographics and Operational Insights



7: Customer Loyalty and Strategy

6. Dashboard Page-wise Explanation

Page 1 – Business Snapshot

Purpose: High-level overview of business health

Key Visuals & Meaning:

- Total Orders, Total Revenue, Total Restaurants – Overall scale of operations
- City-wise Revenue Bar Chart – Identifies top revenue-generating cities
- Top Cities by Order Volume – Shows demand concentration
- Orders by Day – Reveals weekly ordering behavior
- Revenue Over Time Line Chart – Tracks business growth trends

Insight:

Major revenue is concentrated in metro cities, with higher order volumes on weekdays.

Page 2 – Customer Analysis

Purpose: Understand customer profile and spending behaviour

Key Visuals:

- Average Age & Income KPIs
- Revenue by Gender Donut Chart
- Occupation vs Revenue Scatter Plot
- Income vs Spending Bar Chart
- Family Size vs Orders

Insight:

Young working professionals and medium-income groups contribute the highest revenue.

Page 3 – Restaurant Performance

Purpose: Evaluate restaurant efficiency and popularity

Key Visuals:

- Average Rating Gauge
- Top Restaurants by Revenue
- Cuisine Count Table

- Ratings vs Orders Scatter Plot
- Geographical Rating Map

Insight:

Highly rated restaurants generate more consistent revenue and higher order volumes.

Page 4 – Food & Menu Trends

Purpose: Analyze food preferences and menu performance

Key Visuals:

- Veg vs Non-Veg Donut Chart
- Top Cuisines Treemap
- Top Foods by Revenue Bar Chart
- Popular, Cheapest & Most Expensive Items

Insight:

Veg items dominate orders, while North Indian and Chinese cuisines lead sales.

Page 5 – Ordering Insights

Purpose: Understand order behavior and volatility

Key Visuals:

- Average Orders per Day Gauge
- Order Volatility Index Area Chart
- Revenue by Order Frequency Segment
- Cuisine vs Order Value Category

Insight:

Medium-frequency customers generate maximum revenue, highlighting retention importance.

Page 6 – Customer Demographics & Operational Insights

Purpose: Demographic and regional analysis

Key Visuals:

- Age Distribution Funnel Chart
- Cost vs Rating Scatter Plot
- Revenue by City Treemap

- City-wise Orders and Ratings Table

Insight:

Most customers fall within the 20–30 age group, showing strong youth engagement.

Page 7 – Customer Loyalty & Strategy

Purpose: Strategic decision support

Key Visuals:

- Average Menu Price Gauge
- Revenue Drivers Analysis
- Total Menus by Price Histogram
- Orders by Year Line Chart
- Cuisine Ranking Over Time

Insight:

Balanced pricing and popular cuisines are key drivers of long-term customer loyalty.

7. Learnings & Skills Acquired

7.1 Technical Skills

- Power BI dashboard design
- DAX calculations
- Data modeling and transformations

7.2 Analytical & Problem-Solving Skills

- Trend identification
- KPI interpretation
- Business insight generation

7.3 Soft Skills & Team Collaboration

- Structured reporting
- Stakeholder-oriented analysis
- Presentation skills

7.4 Domain Knowledge

- Food delivery business metrics

- Customer behavior analytics
- Revenue optimization strategies

8. Testimonials from Team

- Team members provided positive feedback highlighting the effectiveness and clarity of the dashboard:
- The dashboard was appreciated for its intuitive navigation and clean visual design, which made insights easy to interpret.
- Team members found the page-wise structure helpful in addressing specific business questions without overwhelming users.
- The integration of customer, restaurant, and menu data was recognized as a strong aspect of the project.
- Stakeholders valued the actionable insights, especially related to high-revenue cities, popular cuisines, and customer segments.
- Overall, the project was acknowledged as a well-executed analytical solution suitable for real-world business applications.

9. Conclusion

This project successfully demonstrates the power of business intelligence and data visualisation in transforming raw, multi-source data into meaningful insights.

By integrating five complex datasets and applying systematic data transformation, modelling, and visualisation techniques, the project delivers a comprehensive analytical dashboard that covers all major aspects of a food delivery platform.

Key achievements include:

- Creation of accurate and reliable KPIs
- Identification of customers and ordering trends
- Evaluation of restaurant and menu performance
- Development of a scalable and interactive reporting solution

10. Acknowledgements

We would like to express our sincere gratitude to everyone who contributed to the successful completion of this data analytics team project.

First and foremost, we extend our heartfelt thanks to our mentor for their continuous guidance, valuable feedback, and technical insights throughout the project. Her mentorship played a crucial role in shaping our analytical approach, improving the quality of the dashboard, and ensuring alignment with real-world business intelligence practices.

We would also like to thank Infosys Springboard for providing the necessary academic environment, infrastructure, and learning resources that enabled us to apply theoretical knowledge to a practical, industry-oriented problem.

Our sincere appreciation goes to all team members, whose dedication, collaboration, and coordinated efforts across multiple milestones ensured the timely and successful execution of this project. The division of responsibilities, mutual support, and continuous knowledge sharing greatly enhanced both the quality of the analysis and the learning experience.

We are grateful to the Power BI development team and the broader data analytics community, including official documentation and online forums, which helped us overcome technical challenges related to data transformation, modelling, and visualisation.

Finally, we would like to acknowledge the support and encouragement from our peers and families, which motivated us throughout the project journey and contributed to the successful completion of this work.
This project has been a valuable collaborative learning experience, and we sincerely appreciate the collective efforts and support that made it possible.